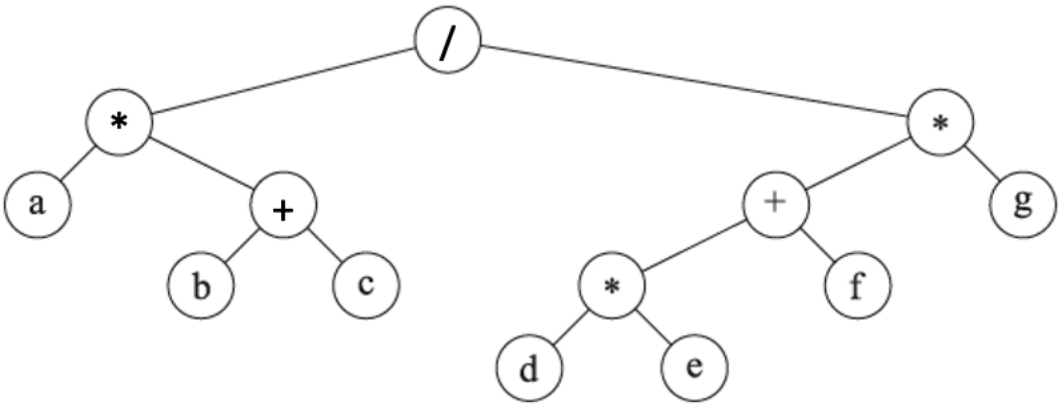
- Please save the file as your complete name and use it to answer the questions.
- Give a short and complete answer after each question.
- When done, convert your file to PDF if possible.
- Upload it to Blackboard using the link in the exam.

- Email me (aelnasan@umassd.edu) a copy of your solution file.

1. Give the prefix, infix, and postfix expressions that represent the tree below: **(10 points)**

(Please pay attention to the precedence of math operations and add parentheses when needed)



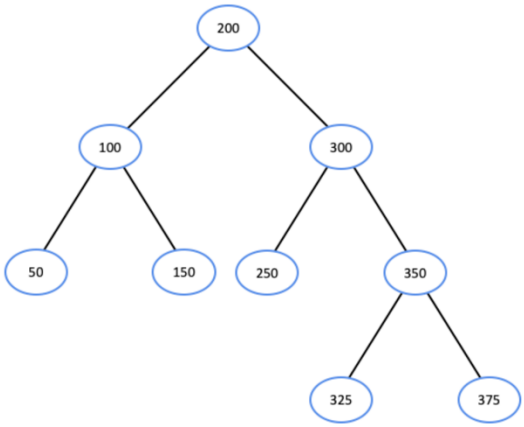
a) Prefix: / * + B C A * + * D E F G

b) Infix: ((B + C) * A) / (((D * E) + F) * G)

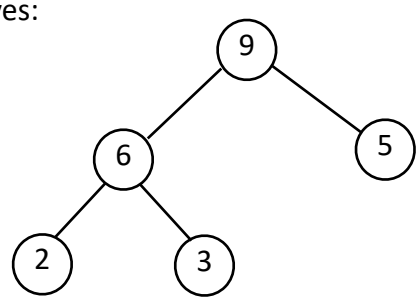
c) Postfix: B C + A * D E * F + G * /

2. For each of the following cases, draw: **(15 points)**

   a. The binary search tree after node 300 is removed. **(5 points)**



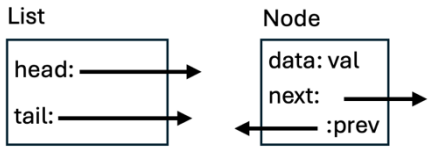
   b. Build a min-heap from the binary tree below and draw the heap after every element that moves: (**10 points**)



3. Construct an AVL Tree by inserting the following numbers: 50, 20, 60, 10, 8, 15 into an empty AVL tree. Draw the tree after inserting each individual item: **(10 points)**

See snapshots for questions 2/3.

4. Using the following structures for a doubly-linked List and a Node, write the code for a function that appends a newNode to someList: **(15 points)**



Please use the space below to write the code (no need to compile the code)

```
ListAppeend(someList, newNode) {

if(head == null) {

    //if list is empty, prev and next pointers are not present
      yet
    someList.head == newNode;

    //assume if head is null that list is empty and tail
    //is also null
    someList.tail = newNode;
}
else {
   //point next pointer of current tail to newNode
   someList.tail.next = newNode;

   //create a temp placeholderNode for current tail
   //not defined in constructor class so doing it locally
   Node<AnyType> tempNode = new Node<>();
   tempNode = someList.tail;

   //assign new tail as newNode
   someList.tail = newNode

   //finally, assign the new tail's (newNode) with the
previous pointer of the previous tail stored in tempNode
   someList.tail.prev = tempNode;

  //current tail's next pointer should be null.
   }

}
```

5. There are six computers: A, B, C, D, E, and F. The table below shows the distance **(10 points)** between each pair of computers. Note that "∞" means that it is not possible to set up the cable between the two computers (hence the cost ∞). Your task is to determine a set of cable connections that will connect all these computers together (not necessarily direct connections) and will also minimize the total setup cost.

	A	**B**	**C**	**D**	**E**	**F**
A	0	9	4	∞	∞	∞
B	9	0	1	5	3	∞
C	4	1	0	∞	2	∞
D	∞	5	∞	0	3	8
E	∞	3	2	3	0	7
F	∞	∞	∞	8	7	0

a) Map the problem into a graph model: what does a computer represent? What does a cable represent?

See screenshots

b) Describe an algorithm to achieve your task.

See screenshots

c) Show how the algorithm works for this specific set of computers and costs. Please present major steps and your result.